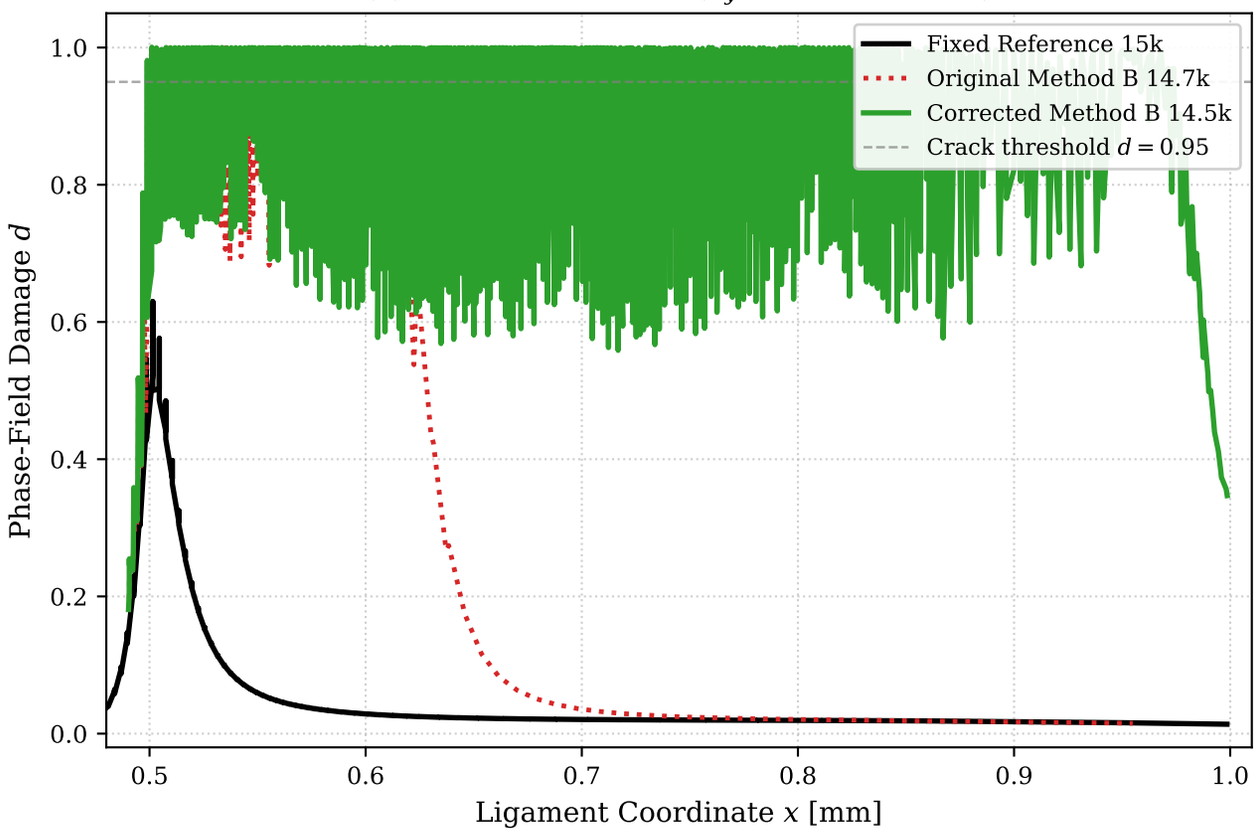
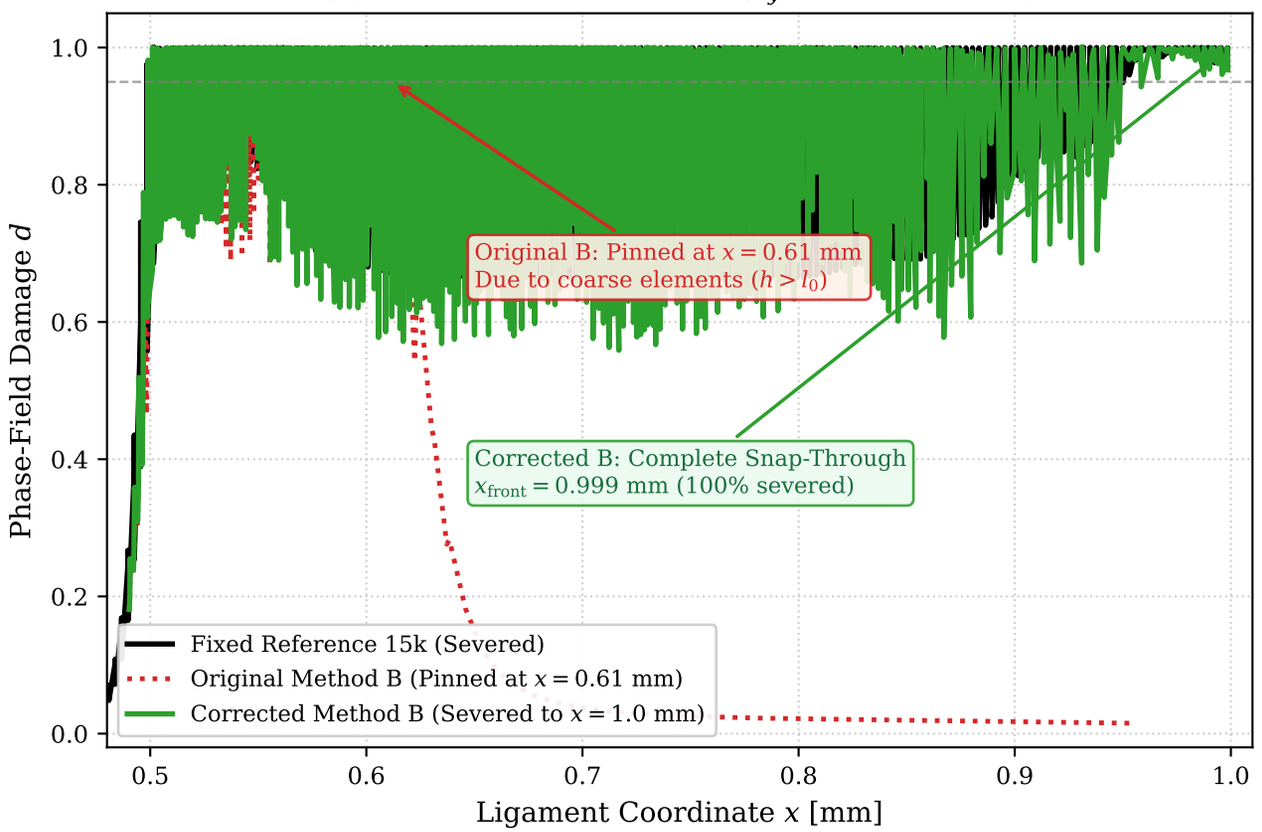
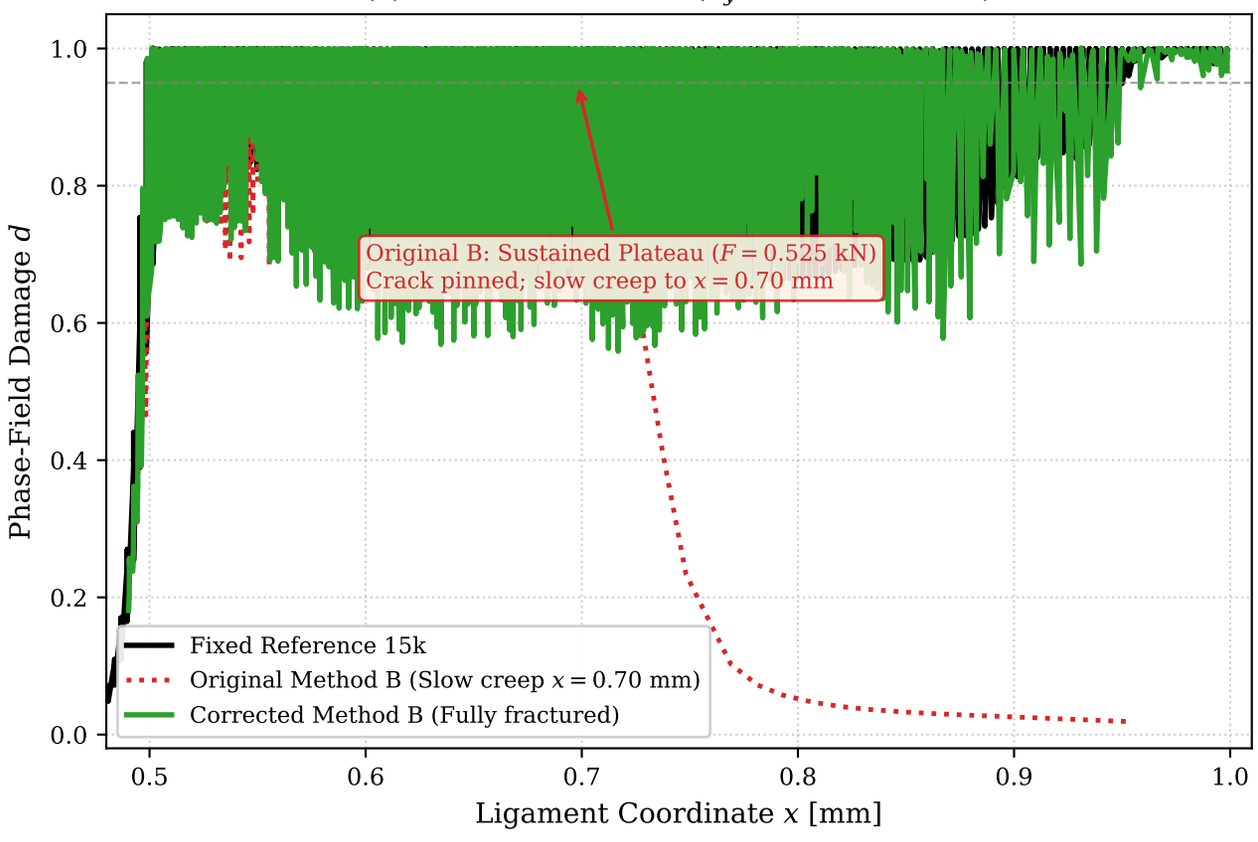
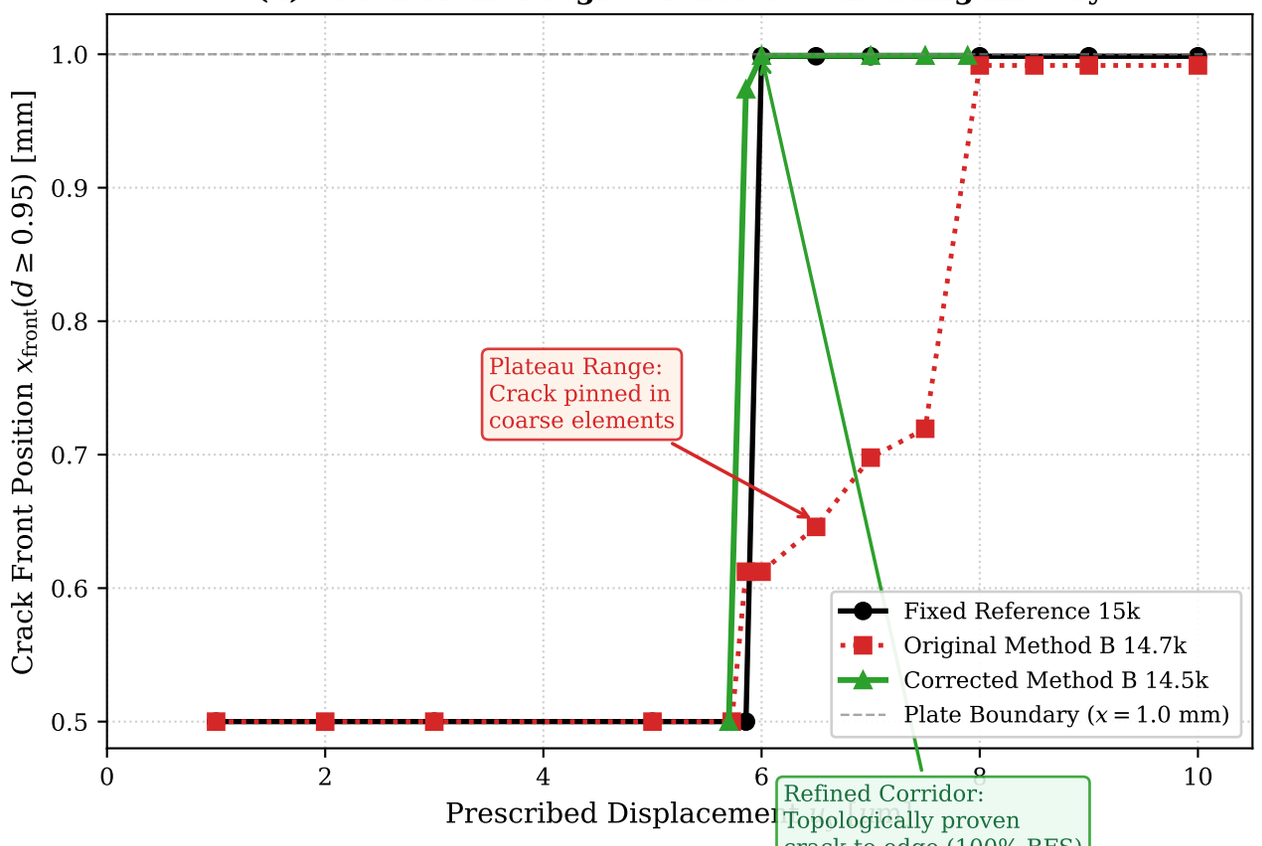

(a) Peak Load State ($u_y = 0.005857$ mm)
Fixed Reference 15k
Original Method B 14.7k
Corrected Method B 14.5k
Crack threshold $d = 0.95$
Phase-Field Damage d
Ligament Coordinate x [mm]


(b) Immediate Post-Peak ($u_y = 0.006000$ mm)
Original B: Pinned at $x = 0.61$ mm
Due to coarse elements ($h > l_0$)
Corrected B: Complete Snap-Through
$x_{\text{front}} = 0.999$ mm (100% severed)
Fixed Reference 15k (Severed)
Original Method B (Pinned at $x = 0.61$ mm)
Corrected Method B (Severed to $x = 1.0$ mm)
Phase-Field Damage d
Ligament Coordinate x [mm]


(c) Plateau Horizon ($u_y = 0.007000$ mm)
Original B: Sustained Plateau ($F = 0.525$ kN)
Crack pinned; slow creep to $x = 0.70$ mm
Fixed Reference 15k
Original Method B (Slow creep $x = 0.70$ mm)
Corrected Method B (Fully fractured)
Phase-Field Damage d
Ligament Coordinate x [mm]


(d) Crack Front Progression Across Loading History
Plateau Range:
Crack pinned in
coarse elements
Refined Corridor:
Topologically proven
crack to edge (100% BFS)
Fixed Reference 15k
Original Method B 14.7k
Corrected Method B 14.5k
Plate Boundary ($x = 1.0$ mm)
Crack Front Position $x_{\text{front}}(d \geq 0.95)$ [mm]
Prescribed Displacemer